

Vulnerability Scan

1. Set up a free trial account on any major cloud provider (AWS, Azure, or Google Cloud) and deploy a basic virtual machine (VM) using default security settings.

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1. Log in to the AWS Management Console.
 2. Open the EC2 Dashboard.
 3. Click Launch Instance.
 4. Configure the instance:
 - Name: Vulnerability-Scan-VM
 - AMI: Ubuntu Server 22.04 LTS (Free Tier)
 - Instance Type: t2.micro
 - Key Pair: Create or select an existing key pair.
 - Security Group: Use the default settings (allow SSH on port 22).
 5. Click Launch Instance.
 6. Wait until the instance state changes to Running.
2. Use an open-source vulnerability scanner like OpenVAS or Nikto to scan your deployed cloud VM for security vulnerabilities and save the scan report.

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- 1) Install Nikto:
 - `sudo apt update`
 - `sudo apt install nikto`
 - 2) Run the scan:
 - `nikto -h http://<VM_Public_IP>`
 - 3) Save the scan report:
 - `nikto -h http://<VM_Public_IP> -o nikto_report.txt`
- **Result:** The vulnerability scan is completed and the report is saved as **nikto_report.txt**.

3. Identify at least three vulnerabilities from your scan report, and for each, document the vulnerability, its potential impact, and a recommended mitigation strategy in a table format.

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Vulnerability	Potential Impact	Recommended Mitigation
Missing Security Headers	Increases the risk of attacks such as clickjacking and cross-site scripting (XSS).	Configure HTTP security headers such as X-Frame-Options, Content-Security-Policy, and X-Content-Type-Options.
Directory Listing Enabled	Attackers may browse server directories and discover sensitive files.	Disable directory listing in the web server configuration and ensure each directory has a default index page.
Outdated Web Server Version	Attackers may exploit known vulnerabilities in older software versions.	Update the web server and operating system with the latest security patches.

4. Pick one vulnerability from your findings and implement the recommended mitigation (for example, update a package, close an open port, or adjust a firewall rule), then rerun your vulnerability scan to verify if the issue is resolved.

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- 1) Selected Vulnerability
- 2) Directory Listing Enabled
- 3) Mitigation Steps
 1. Open the Apache configuration file.
 2. Locate the directory configuration.
 3. Change:
 - Options Indexes FollowSymLinks
 4. Save the file.
 5. Restart Apache:
 - sudo systemctl restart apache2
 6. Run the Nikto scan again:
 - nikto -h http://<VM_Public_IP>

Result

The second scan confirmed that the Directory Listing Enabled vulnerability was no longer detected, indicating that the mitigation was successful.

5. Share your vulnerability scan results and mitigation steps with your group using a shared document or chat, and discuss how these issues might appear in popular apps like Zomato or Paytm if left unaddressed.

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- The vulnerability scan identified several common security issues, including missing security headers, directory listing, and outdated software. After applying the recommended mitigation for directory listing, the issue was resolved in the follow-up scan.

❖ Discussion

- If vulnerabilities like these remain unaddressed in applications such as **Zomato** or **Paytm**, attackers could exploit them to access sensitive information, expose internal files, or compromise customer accounts. Regular vulnerability scanning, timely patching, and secure configuration help protect user data, maintain service availability, and reduce the risk of cyberattacks.